

ActInf Textbook Group ~ Cohort 5 ~ Session 18

(Chapter 8, Part 1) ~ 3/4/2024

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all right welcome back everyone we're in the first discussion for chapter 8 so does anyone want to begin with like just any thought or question on eight any quote or or part and we'll just see where it goes H yeah I was I'm was curious about the role of attractors um in the yeah how how the the role of attractors in the um generative let me let me try to find the exact quote section yeah just basically the the Ro of fixed point attractors in in the generative process anyone with a thought on that all right there's a lot to say about a tractor behavior in dynamical systems overall and that's kind of where we're at one way to think about it is if there wasn't some kind of attractor point or set or limit cycle or something like that that variable would like never converge to a stable stationarity so if something's oscillatory it could be thought of as like converging to a cyclic attractor if something is kind of like dampening like a spring that's like a fix Point attractor plus dampening or it could be like just oscillating around a fix point of tractor so it's like dynamical systems are of certain types analytically and they can be converging and dampening converging oscillating limit Cycles all these are like different outcomes or possibilities for dynamical systems but to kind of pull back um a step so chapter 8 follows on chapter 7 which shows the discrete time formalism so in chapter 7 discrete time steps in the simulation are explicitly modeled like we have time 67 time 68 Etc chapter eight is going to show a totally different complimentary treatment of time for generative models which is the continuous time formalism in this chapter is about like unpacking a lot of the differences from the discrete time and one of the key differences is like a different take on the relationship between observations and hidden States so in the um discrete time setting and figure 4.3 is like the Rosetta Stone with the top discrete time model and the bottom continuous time so in the partially observable markov process setting the a matrix is kind of going both ways between the hidden State and the observation here kind of in line with the continuous time formulation being like uh closer to some Physics based uh formula there's there's two core uh equations there's observations y same as observations here and then $\dot{x}$ so the um derivative on hidden

States so derivative on hidden States here and basically each of these are mapped out so there's like a kind of regularity of observations or a function of observations and a noise function and then there's also an underlying generator and this is kind of coming from Friston and at all's earlier work with SPM where they are like all of the sensor data for the neuroimaging like fMRI or EEG and then $\dot{x}$ is the rates of change of the underlying neural activity that's related to the observations that are coming in so that's kind of the that could describe any G and any F so then they by 8.3 get to a kind of simple generative uh model where there's a fix Point attractor so here's the what it looks like to have a fixed Point attractor so this is basically when X is less than V the set point the rate of change is positive so it goes up and then when X is higher than the set point the rate of change is negative so it goes down so it's kind of like a first order thermostat with a set point encoded in the underlying Dynamics like this any thoughts on that or does anyone like want to add more pieces to this about attractors or anything yeah I think um the the attractors become like significantly more interesting whenever we move into the the next uh sections with like the the luck volterra Dynamics and the lawen system just like trying to map them to like other phenomena like I really like the um the bird song example I guess this is around 100 page 163 um where we're introduced to the concept of generalized synchrony and so the birds are basically like their internal states are starting to synchronize as they're like trading off singing with one another and trying to kind of match or continue the same song so to speak and here we end up with this kind of like like the states are orbiting around in that lower left figure two different I suppose attracting points so it's like the state is like orbiting between the two um but then my my next question is I'm I'm not sure exactly what those two orbits kind of represent if they represent something like one Bird versus another or if this is like the state that one of the birds as a single agent is like its beliefs are moving around you know its own um actions or or beliefs uh versus like whenever it's not singing and that's uh my next thought yeah that's a good question or or I mean a good uh like progression how first we have a simple fix Point attractor so this is not even a moving fixed point just a set fixed point then kind of calling back to chapter five there's a descending settable fixed point so that was in the the cross-section of the spinal cord where there's the descending motor prediction and then the proprioceptive data coming in and then so it's kind of it's the same exact mechanic as the um set point it just with the additional piece that the set point is uh a descending prediction then um another bump up in what the attractor can be going to from stable fix Point moving fix Point um now this gets into the kind of like winnerless competition or like generalized dynamical the generalized sequential Dynamics these are like iterative

processes and then first there's the um this is kind of like a uh non-chaotic winterless competition with an ecological fluctuation so lockable Terra arose from the ecological Predator prey modeling which is kind of what is referenced here plant herbivore carnivore but also those have been applied extensively to like activity patterns of brain regions so if you have or or Gene regulatory networks like if you have multiple components that have um s features such that when they're low they can ramp up but then when they're high they're like self-limiting directly or indirectly then they go up when they're low and then they go down when they're high and so they stay within like a bounded interval but this is still has a kind of a very regular um movement so it's three variables that are linked in a dynamical equation p h and C in this case plant herb carnivore and then here are like two projections onto a side of the cube and then that gets taken into like one more level to go from the non-chaotic dynamical into the deterministic chaos which is just kind of alluded to and explored more in the citation but [Music] then that takes us to the turn taking Susan so so I'm I'm wrestling with um you the attractor States in complexity science and and um chaos theory and so I'm kind of wondering if that's the steady state is you know is the the bounds in other words you you're going to move um you know it's about how you're calculating the movement and although I you I don't want to might be better to put this conversation um in the next after the next chapter but um but in terms of you calculating in it may be that it's just a different nested model of what is the current state in other words in other words are is this this state is going to look different if it's if it's over capacity it's out of a state of resilience so I'm assuming that that's actually another nested model and I don't so I don't want to try to combine the two or am I thinking about it incorrectly yeah it's interesting like explore it for specific system and then maybe it'll be clear like which aspects of resilience are accommodated in the model like as written like even the fixed Point attractor I mean in a sense that's a rudimentary form of resilience because when it's pushed away from the fix point it comes back to it but there's multiple resilience Concepts like one of them being that it returns to where it was before another resilience concept is that it stays where it's adapted to so those are kind of not even compatible definitions of resilience which is like why when you build a specific model then okay for this variable when we say resilience we're talking about the adaptivity like definition versus like the resistance to change version and then it's sometimes really interesting how even some of the simplest formalisms can capture aspects of those Dynamics but then like you can't use the model to show something outside of the model so there if if this is the only one you have you'll be able to have that kind of like first order um fixed Point Behavior but then it wouldn't be adequate to

describe some other kind of behavior like this one this kind of model will never show a moving set point but this model would but it wouldn't show chaotic Dynamics but then this one would so head's trying all these different kinds of models and just like like in the textbook which is being shown here just showing all these different features okay then from the chaotic to the chaotic with learning so it's just all of these different patterns and then it's kind of like well for the phenomena for the system that you're looking at which one of those like levels or types of um Dynamics feel appropriate yeah and and and just to put context around this this is just trying to get give us the basics you know in other words not to throw in too much more complexity I mean once we understand the basics then we can you begin adding more degrees of complexity I guess Thomas Parr in his um book stream gave a really good kind of timeline talking about how how how um the earliest implementations were the continuous State space then people started thinking about getting sequential Dynamics this is exactly what we just saw with lock of voltera then people wanted to look into planning and sequential Dynamics with like counterfactuals so then that motivated more of the discrete space discret State space discrete time model to to get that kind of decision- making and planning um then there were hierarchical models that were able to Nest either discr or continuous or Hybrid models um eventually by being reintroduced introducing the continuous State space as the kind of lower sensory motor model and then having like a kind of higher order discreet model and then that was kind of explored in the folk psychology work where we have continuous time and state spaces like for sensory and motor and then you have discreet State spaces and time for more sequential decision making and planning Susan I think you just answered it I was wanting a little more definition around discrete space and continuous space but anything else yes discrete State space is where the options are finite and discrete like one or two or three or four or 0.1 or Point 2 or 3 but discrete clickable values versus The Continuous State space is when there's a Continuum that's like differentiable like at all points you could put a ruler and determine the slope whereas if you only have one two three four you can't really like take a ruler and find the derivative at each time step like you can't you can't really have the derivative in the same sense for a staircase it's like a step function so it's true The Continuous and the discreet state spaces um we could be talking about temperature like temperature is a continuous variable temperature is a discrete variable and then one special variable for dynamical modeling is how time is addressed and so time as chapter 7 and eight Explorer time can be explored as a discrete State Space chapter 7 or time can be explored as a continuous State space like chapter eight Andrew yeah just just like the way I kind of think about um the whole like discret space

model on the uh on the higher level of a hierarchical model and then continues at the lower um and I I don't know if this stands up to anyone else but um something like um I'm sitting here I'm trying to figure out if my heartbeat is too fast like yes or no to discrete options I have to like pay attention to my heart rate for some time but of course like I can't just go with a single time step I have to see like you only know the rate by you know taking a a second or a few seconds to figure out what the rate is and so over you know however many seconds that I need to measure that which is multiple time steps or a faster time scale I can then make the discret decision if it's too fast or not if that makes any kind of sense so thus yeah and that's a perfect um figure 8.6 that that um Daniel pointed to is like we see how there are multiple time steps in that continuous level at the lower level U for every single time Step At The Higher One um and so yeah I think when thinking about modeling like say an agent and trying to include things like physiological uh signals like heart rate and other things like that there's this kind of faster time scale going on and then the agent at a higher level maybe a cognitive level or some such thing is making like more discret decisions or or inferences based on what's going on yeah he anyone else have just any any thought or or question on eight we can look at some questions we can continue but the chapter's very compact it just moves through those sequence of examples fix Point attractor fix Point attractor moving point attractor dynamical attractor chaotic attractor with learning then it just says and you could put that continuous model at the lower level of a discrete Time model and here's an example of what that looks like like at the higher level is which location to fix it upon and then the continuous lower levels like the Cade movement and then there's a few settings where that's been studied and that's the whole chapter J uh I have a little question about the small question about this bird uh system with the chaotic attractors or chaotic Behavior so if I remember right then chaotic Behavior Uh the by definition it means that you can't that you only have a limited Horizon where you can predict what is going to happen uh still these birds seem to be able to synchronize uh and is it if I look at the figure 8.5 does it mean that they do not synchronize their down to the right that they even after learning they don't synchronize perfectly they sometimes lose each other but then they uh but they stay quite close like they they because if you can't predict what is going to happen you probably can never because you don't know the initial conditions you know you never know exactly where the other the other individual is I guess but is that correct or yes so one thing to note um as uh as I think it'll probably say um singing from the same hym sheet so just to be clear this model has presented it's not like this is like a generalized model of semantic communication and turn-taking in conversation this is like two birds with a fixed

innate song that they can kind of in the second part of it do some local tweaking but in the first part they can't at all but they have the same song template so they're not engaging in more like open-ended um semantic transmission it's kind of like like a chant that's like supposed to be ongoing and so the only options that they have are either like listen and understand that the other one is singing it or notice that that has stopped and then take action to to bring that song like back into play that's kind of like the situation that the bird song is modeling um but then they show like in the paper that when you have one bird with a precise song and one with a very like imprecise song that that the like that younger imprecise one can like learn and converge to the the other song then you can kind of fix that Bird's precision and bring in a new plastic one and then converge that one to it and so that got into like the kind of inner generational transfer again of like a templated song with a little bit of Drift But yeah what's Happening Here is like between bouts of singing There's this kind of tightening of the joint Dynamics so this could be like if this this would like two people and they're pointing at each other pointing and they're trying to they're doing an improv game where they're like trying to keep their fingers as close to each other as possible um so um if it the movements were perfectly correlated then they would be on like a onetone line like the x value of one would be matched exact tracked perfectly by the x value other that so and then if they were totally uncorrelated it'd be like a perfect circle which is say as wide is as it is tall so like the correlation would be zero there wouldn't be a significant regression through that cloud of points um so they start in the same ballpark because they have a template and then there's some fine tuning in the learning process so that their expectations tighten up it's like and then the turn taking part and then so about the chaos and predictability so that's where the um leap exponent comes into play which is kind of like this variable that describes how chaotic a system is ranging from a system that's not chaotic and the kind of metaphor there is like a laminar flow so you have two planks of wood on a laminar parallel flowing river those two planks will stay near each other at the same distance and then there's other rivers where um planks that are placed near each other converge and then there's systems where planks that are near each other diverge and that's chaotic because you can't necessarily know it's like you're coming in on this one and then if you come in just on this side you take another attractor to the right but if you came in a little bit more on the left then it's spins out and so then as that exponent gets more uh more chaotic it gets harder and harder to predict points um trajectories of points that start initially next to each other so the horizon predictability drops and so this is kind of like it's like this one is saying like we're on we're on the right side of the turn taking

and then it's like the Precision is dropping and that's kind of spiraling it out and then eventually it spirals out and then it flips to into the other attractor spins in spins out and then eventually it flips out there just showing that those kinds of switching time Dynamics can be modeled even with deterministic equations so there aren't even like this isn't necessarily even stochastic this is just the deterministic chaos which was kind of like that's one of the most interesting and exciting parts from complexity theory in chaos is like single pendulum is deterministic and non-optic but then the double pendulum is deterministic still because it's just two single pendulum but then it has the chaotic Dynamics also connected to Wolf from some computational irreducibility and about how like there's things where even if they're deterministic that it still can't be determined like if the program's going to do a certain thing or finish at all yeah Susan so I guess that would have a lot to do with the policy selection and is policy selection would that would that be the same as you know what model you chose to you know to you assess a situation yeah like the singing Behavior just on the first level the singing Behavior can be understood as the policy the motor policy of the bird okay whether it SS or not like in a in that binary system um but then a little bit more cognitively the choice of which framework to apply or which interpretation to apply is like a mental policy decision so so where would affordances come in it's a good question in um the of reference live stream 28 with Mar sanit Smith so this is kind of a different this is another step by step so here's static perception hidden State sensory ambiguity to the observations D prior now in the textbook they went from that kind of static perception and then went into the dynamic perception but here there's kind of this intervening step with a Precision variable on a so this is like changing the temperature or the prec the Precision on a which is to say from sharpening it so that it looks more like just ones and zeros and there's a less ambiguous mapping between hidden States and observations all the way in the low Precision extreme a is just kind of like blurred out so that observations have no information on hidden state so they're going the attention as Precision here so now we have kind of the discrete time figure 4.3 or figure 8.3 whatever it happens to be with these little Precision guys on the bottom but otherwise it's the same and then let's say that this blue is the sensory motor layer so that could be the um spinal cross-section or that could be the kind of gross motor Behavior like singing here's where the mental action or the internal affordances the covert action is coming into play where where that affordances in that nested level are modeled as basically Shifting the Precision on the lower level and then they had a third level of policy about that second level policy but everything past the blue sensory motor layer are just higher mental affordances oh very cool yeah that

was kind of a big moment when the same or similar kind of um formalism that was used for sensory motor control and affordances externally got brought in to this kind of internal attention as action setting by by Lars and and others in 2021 and then they still like even in the previous days continue to like make this more comprehensive and like just bringing in more pieces and bringing in like World Knowledge traditions in a really interesting ways too thank you yeah in discreet time I don't know if there's any continuous time versions of this but that would be cool too yeah Kristoff yeah I just wanted to make a comment that um I think it's interesting that this formalism can actually describe both um you know continuous time models and discrete models because as we as we move in in in the hierarchy of Concepts then we finally arrive at at things that are discrete but discrete phenomena in some sense are not um um you know there there's there's always this uh this sort of uh problem of course graining from um from underlying processes so there's the generative process that gives rise to the phenomena we are sampling that in some way and our we're computationally bound so we can't have perfect model of the other system so we have to sample so that gives rise to some kind of discretization of uh of the phenomena and temperature I think it's it's an interesting example where you know we can model it as a continuous variable but it's not actually useful um to our um uh for us temperature always has some kind of discretization because we don't care about I don't know six seven eight decimal places right so so the fact that you can build this hierarchy and this um um uh discretization just kind of naturally fits on you can fitted on top of continuous models is actually U very reassuring that this is a this is a really good Paradigm that these two things can be connected um kind of directly without uh without bringing in some um some other Theory to to say how these how these systems would interact um because that's just kind of what naturally happens uh in in more complex systems at the top level things do look more and more binary uh when the decision- making is made do they go left or right in the real world there is no really left or right there's just there's some kind of sensitive Direction but um at my at my mental level that is that that that might be a binary Choice yeah very interesting thank you um Kristoff or sorry um um your hand was still raised on my computer but did Pros did you raise your hand uh uh yes uh actually I was uh Christoph actually said what I was trying to say uh like I was thinking like any kind of complex system will have lot of continuous uh items in the downstream but as you go up there's like a decision making happening and uh fin it's it's more binary but it's like so the the the key learning which I'm get getting over here is that uh uh like there are various possibilities in which you can mix and match this kind of uh systems that you want to have but there is a possibility in which you

can there's a flow and we can use that for our own purposes yeah that's very interesting another kind of topic to connect this to non-active um psychophysics is the just noticeable difference concept so it's like um the classic example is like we have two objects that are going to get put in one one in each of our hands and then the task is which one is heavier so the objects have a continuous value value in reality or you could think of it as your muscles are exerting a continuous amount of tension to prop up and like a value the um mass of the object but it's a binary decision which one is heavier um so if the stimuli are exactly matched like if they're the same weight but there's no option for the the same weight you have to say which one you think is heavier but you'd expect that 50% of the time like that it would be guessed which one is heavier and then it turns out that there's like this kind of sigmoidal curve where there's a high sensitivity when things are very close to each other to differentiate them but then if it's like 5 pounds versus 15 pounds changing the 15 to a 16 it's not like it increases the decision-making accuracy by 10% because it's already at like 99.9% so that's kind of an interesting thing is like connecting this the stimuli intensity to like potentially the binarized decisionmaking sigmoidal described decision making around that just noticeable difference um Point Kristoff uh yeah there's also very interesting research on that on that subject um uh called qualia Quantum qualia they they're essentially testing a hypothesis that the the quality of experience depends on the history of experience so it's not that uh when you perform this kind of experiment say asking people um uh which colors are uh more similar and more different depending on what the sequence of color is that they've seen already their responses um are likely to be different so there are some people trying to test theories that essentially bring in um you know some kind of mathematics that the kind of mathematics that's uh previously been used in in quantum theory to model uh model phenomena almost like like a evolution of a of a of a you know Quantum wave function so that when when you're performing an experiment it it really matters what what what your context was and there's some some really interesting results in that um that space so uh there might be you know it might be that there is sort of like a reset time that's um that's required but it might be that simply that um we we we really perceive deltas and that's what really matter so if you give someone two very heavy things and then to ask them to compare two light things they may just their brain might be used to feeling the feeling of something being heavy and not having enough Precision not resetting enough not having the Precision to perceive the difference between the the the lighter things so it's really really interesting how um um how that works yes yes and definitely like about detecting change like the eye is expecting a certain intensity of

light and then it's brighter than expected that induces like a dilation or or a constriction however it goes and then that recalibrates to a new expected set point and then from there again it's just about the direction of like whether this is less or more that's kind of the predictive processing Paradigm which is like you don't have to always be carrying through like the temperature of the room if it just happened to be that your implicit set point was room temperature and then you had that fix point of tractor all you'd have to carry through would be like plus or minus within within reason and it wouldn't need to be like 72.4 and do some sort of computation and then store that you know you just have a binary decision about which is action dependent yeah so it seems to be that like the bias seems we we always continuously adjust the bias to to to really see just the differences that make the difference and I think there's a really fantastic uh you know uh example of this from Antiquity one of the Greek philosophers um I forgot which who it was but he was staring at the uh at the river and um seeing people just trying to uh I think rescue some animal that just went into the river and and so on and then and then he looked at the at the ground after and he saw the ground flowing so so even if it's not like a physical um phenomenon like you know expansion of the veins to pump more blood or whatever um even just at the level of the brain there is some kind of processing that you can very easily see yourself if you do it if you look at something that flows like a water flowing downwards on the TV and you look at the wall you will see the wall flowing because the brain adjusts to that flow so that you could pick up the differences within um you know the flow not the not the flow itself you just become oblivious to it um I wonder is there any is there any work in um active INF that would try to model something like this because that that seems like a really interesting project I don't know specifically it reminds me though also of like going on a long walk or hiking and looking at the ground a lot and then stopping and then looking far away and they're kind of like being like zooming because there was even like a calibration to the um the amount of visual flow or just like an expectation that like objects come closer or go further or something like that and like um another fun example like the Exploratorium in San Francisco it's like fun like Hands-On science was like a pipe that had um a colder a hotter and a room temperature component and then you put like one hand on the colder and the hotter part not burning or freezing but just noticeably and then you put both on the same section and then like one of the hands felt the one that had been on the hot felt like was cold and vice versa but of course you're looking at it and your hands are on the same piece of metal but each hand was only able to kind of like report that it was it was getting contrasting opinions so felt like as if you had flipped the hands but they were both on the same

temperature and after images visually some of these phenomena like are more at the cellular level like some visual after images and effects those are like it's like some you know retinal cells are exhausted and then like they so it's like that that phenomena I mean inquiry finds that that phenomena can be explained by cell level findings whereas something else like a ballerina that rotates or a cube that's collapsed one way or the other like those are not going to arise at the level of like the single retinal firing rate but they might arise at some attentional level yes Andrew or anyone what what what is quote what does the quote mean or how does it start off the chapter oh I just threw that in there in reference to what Kristoff was referencing uh I was I'm not sure if it's heraclitus who saw the um yeah who saw the animals go into a river but I know that this is a famous quote referencing like yeah there's another line like no one steps through the same river twice like you're you're observing like you're observing behavior and so everything is continuous but I don't want to pull us too far away from uh from the chapter with just saying that but I mean I clearly to me it seems to just be representing like continuous Dynamics right continuous change over time so yeah totally Pros no I was just uh trying to say a little bit about the quote I think like uh as far as I remember it's like you cannot step on the same river twice because uh after a moment the river has changed you know like it is not the same river it's like the concept like verbs like it's like everything is changing and it is not the same so yeah just just a find yeah like so when there's like a rock in the river and then there's like a flow over it but the flow is kind of like kept at a c like there's a stable point for the flow that's kind of like non-equilibrium steady state because there's still pass passing through a material dissipation of a gradient but then if you were just studying like the elevation of the river at that point it would be a stable or it would have like a little ring around a small point just on the kind of like practical note um pmdp the main python package for active inference generative models is discrete time only hence the PDP tutorials are like all discrete time whereas Sanje's work in Python accommodates continuous time models as do the um matlab earlier SPM work and RX and fur and Julia but may maybe someone has brought some continuous time models in the PDP I just don't think it's on like the main branch or or I'm not sure if it's on any branch but it's definitely in sanje's code SPM and in RX fur the papers that are cited like at the end um we could double check but just looking at um the the authorship in the Years these papers are probably almost all in matlab we can double check them but very probably continuous time python Pros um ask sanje just in Discord just tag sanje um is this wait Andrew is that a continuous time well I just sent for the general documentation but yeah I mean all the examples are there yeah yeah this one's a discrete time discrete time and discrete

State spaces actually the this one has discrete and then the continuous State spaces but it is it all it is all in discrete time but if you um in Discord maybe ask sanj Andor like tag Magnus kudal and sanj and you'll get to get answers okay thank you yeah yeah um like this this be fun collaborative annotating for whoever wants to do so like with with all of these examples and more that we can find um we could annotate like time discrete continuous either or like hierarchical and then understand which one of the code examples as we also continue to add them has these different features if anyone wants to explore that we can definitely do that but right now it's just like the links but but maybe look through them and see which um which ones have which features yeah Christof uh yeah just I just checked my sources and um the the story I was referring to was um about Aristotle so apparently he saw a horse stuck in a flowing river and he fixedly watched the rescue operation and he finally looked away um and everything else appeared to flow so he was uh it was the first um uh recorded visual illusion uh in history wow horses kind of play a key role in the visual perception because like the first highspeed photograph was about the debate whether the horses pick up all of their feet when they run and then it like they took a photo and it was resolved like yes when they Sprint all four of the feet are off the ground but that was it was not known till that happened well you know in the first Polish encyclopedia um written in in the Polish language there was a very interesting you know note about the horse it just simply said said horse what a horse is everybody can see don't ask if you don't know but you know it tells you something about the you know about the times definitely there's been a lot of recent work on the time perception and the flow of time like Darius and and others and some some upcoming live streams too have kind of explored that that's definitely a challenging topic because it's dealing with something experiential but people AR pursuing it um whereas these examples like the eye movement that's more empirical um arguably this is information like pupil arousal and and icting this is information that like potentially every smartphone and webcam collects all the time I think this is you know one of the interesting differences between machine and human or machine and biological vision is that um in biological creatures we do have these mechanisms and um of you know adjusting Precision adjusting um resolution of picking up salient uh features whereas machine usually um and we have it a sort of at the low level at the um um the level of the eye that a lot of this processing happens whereas in machine you just get a big array of pixels and the neural network is supposed to um figure that on on its own uh from that input um and you know I've been wondering for quite a long time whether you know things things things are the way they are for reasons and we may not understand reasons uh why things are

the way they are but evolution is not is not stupid it produces things for uh um for for for really for for really good reasons so I I think that it might be that um the fact that we have to sad to to to Really look at something that we have this the all these um all these limitations that actually in in my view leads to more robust uh represent ations um I don't know how to test this hypothesis in in a in a digital neural network yet but it strikes me as as as highly plausible that this limitation is necessary to build um robust representations because robust representations would be the only thing that survives this kind of um sampling process and if you get too much detail you you just get uh too much uh to it's harder to extract information sine information that's very interesting whether there's like more fundamental um like computational limitations that's highly possible but like just when it comes down to the fobia the center high resolution area on the eye it's like there are trade-offs just with the density and the energy costs of having high resolution in the center of the eye and so like here's a visual Cade model and then it's possible for this like icade um agent to to differentiate these different um visual patterns like is it just line is it like parallel lines with this diagonal thing or you know vertical lines those can be quickly differentiated like am I looking at a Dorian or ionian pillar that could be differentiated with like one to three skillful is secades it's like am I reading a scientific paper it's like well is the lcav logo in the top left no okay cic over here is there something there no okay last check okay it's not a paper that's whereas if um this were going to be taken into like a sort of classical machine learning pipeline then let's just say this is like 20 by 20 well that's 400 pixels coming in so you need like at the very least the memory capacity for like that many pixels to be coming in versus only nine pixels with a 3X3 high resolution and then that's like the kind of Truth drop on our visual experience being the generated not the processed like no blind spot color in the periphery resolution in the periphery suppression of the Cades like all those features are the information that does like give evidence in the moment to moment that we're experiencing what we generate visually and then probably by extension other senses too not like taking in a whole whole scene and then distilling the scene but having an underlying generative model of the picture even if it's our first time seeing that specific one and then doing icting to fill in the detail and then doing um all kinds of other cognitive phenomena like attentional modulation and kind of like instantaneous overwriting to just like make it normal make it feel normal yeah this this is uh I guess this will be in one month this has to do with the time perception affect temporality intention retain pass the sedimented retained embodied trace and then the um unfolding kind of time cone counterfactual possibility prospective potentiation space kind of just blurring off in

last minutes any like other thoughts on eight otherwise yeah it goes fast basically that's eight then next week earlier for Gort 5 we'll continue to talk about eight maybe um Andrew if you want to before then um or anyone else like maybe we could look at one of the papers in more detail like the is aade paper with Thomas Parr just because I mean we've kind of covered the chapter in these conversations and we can look at the bird song or or um some other work with uh that's s of the chapter then chapter nine that gets into the empirical data basically up until this point um all the discussion on building the generative model the great time in continuous time has been like building it by Fiat and then simul or making synthetic data but then if you want to run it in the other direction and take in empirical data and then parameterize your model for like determining within or between group differences or doing any other kind of testing like that's where chapter nine comes into play and then chapter 10 just summarizes it up so it's so funny the book's like an accordion like sometimes it seems like there's a ton other times it's just like barely it's just like just quickly like flipping through index cards of all these different models because there's so much detail to go into um with with even the simpler models so so just to confirm nine will go will cover the um spatial boundaries that when you said parameters is you know we've talked about the the time parameters and I'm trying to um trying to Envision you how do you set the spatial parameters that's what Nine's going to cover good good question parameters could refer to spatial or temporal or like cognitive any kind of parameter but if we're going to be talking about data coming from a spatialized system like a room that has ensors in locations then part of making a model to deal with that data would be how do we incorporate the spatiality into this model so if our data have a spatial component I mean if we don't take that into account at the very least we've left important information out about our system that we know worst case scenario we've come to misleading conclusion because just kind of like averaged out all the sensors but they were in different locations you know on the roof so then there's like a really you know the roof is like a triangle and then it was like there's a temperature pattern that makes sense when you know that some are on one side and some are on another side Okay the reason why time gets like such a special treatment whereas like space we could model it continuously like we could have a continuous GPS coordinate it with just like as many levels of precision as we wanted or we could discretize GPS and just say okay we're only going to like the first decimal point and then it's a grid so the reason why that's less of a big deal is that the the time concept gets taken into account every single time we iterate the model forward so how the model changes in time is like obviously one of the key aspects it is what makes it that chemical so that's just a huge decision that's why

like in chapter 6 one of the big decisions that's brought up is like how is like should it be handled should it be discrete or continuous because that it's a huge difference whereas it's a smaller difference if we have um like temperature as a continuous or a discrete value temperature could be modeled continuous or discrete whether we were thinking about time discrete or continuous so it's kind of like even just with the thermometer in the room like no action no nothing like this you have a 2 by two discrete temperature discrete time continuous temperature time off diagonals like and it's kind of like simple examples like that that give good grounding as to why multiple models are constructed because even if it was just a air conditioner in a room you'd have at least those four models at least in principle and then you could explore like which one have which ones have different um which ones explain different am proportions of the variance of the data which ones have different computational resources and okay well this one's more accurate but it's more computational resources okay we'll run that on the special one or we'll run the cheaper analyses first and then if we need to we'll rerun it for the final version with the more expensive you know those are just like practical considerations that come into play when you're working with real data and systems all right thank you see you all next bye